PART 4 - NEW CONTENT ONLY

ERP Revision Notes

Remaining Topics After Part 3

Scope: Only the concepts studied after the previous PDF. Earlier revision-note content is intentionally not repeated.

What this PDF covers

Block	What you need to remember
Functional modules	ERP modules are separate by function but integrated by shared data and processes.
BPR + ERP	Fix the process; then configure ERP around the improved To-Be process.
Readiness, success, risk	People, process, data, technology and project readiness determine ERP success.
5 Cs	Complete, Connected, Cognitive, Compliant, Capable.
Implementation approaches	Big Bang, Franchising, Slam-dunk.
Value Matrix	Business value focus evolved from Production -> Cost -> Customer -> Service.
ERP/vendor selection	Fit, Cost, Scale, Support, Integration; avoid unnecessary customization.
Master Data / Item Types / Signal Code	Core reusable records, item classification, and warning/status indication.

Source basis: Reading_Materials_01.pdf, Reading_Materials_02.pdf, ERP FAQs_updated.docx, and the previous-year questions you shared. Where the course files only name a term but do not define it, this PDF marks the explanation as a **conceptual expansion used in our study**.

1 Functional Modules and Manufacturing Module

ERP module: a part of ERP designed for a business function. Modules work separately by responsibility but share information through one integrated system.

Sales	Inventory	Manufacturing	Procurement	Finance

Module	One-line purpose
Sales	Customer orders, pricing, delivery requirements and order management.
Procurement	Supplier selection, purchase requisitions, purchase orders and buying.
Inventory / Materials	Stock, material movement, warehouse/location and item availability.
Manufacturing	Production planning and control using MPS, BOM, MRP, capacity and shop-floor logic.
Finance	Financial transactions, costing, receivables, payables and reporting.
Distribution / Logistics	Movement and delivery of finished goods.
HR	Employees, attendance, payroll, training and employee records.

Manufacturing Module: the one-line map

Demand	MPS	BOM	MRP	Capacity	Shop Floor

Exam sentence: The manufacturing module supports production planning and control through MPS, BOM, MRP, capacity planning, item control and shop-floor activities while integrating with inventory, procurement, sales and finance.

Item Control - application

Purpose: standardize and track each item through item code, description, unit of measure, category, stock, location and relevant planning information. One item should have one shared identity across functions.

Example: Use RM-HDPE-001 instead of different departments calling the same material "plastic", "HDPE", or "raw material 5".

Phantom Item

Conceptual expansion used in our study: a logical BOM assembly used for grouping/planning that is generally not stocked separately as an independent item.

2 BPR + ERP: Chicken-and-Egg Paradox

BPR = Business Process Reengineering. Redesign the business process to remove waste and unnecessary steps instead of simply computerizing the old process.

Question	Risk
Do BPR completely first?	The redesigned process may not fit the ERP; heavy customization may follow.
Do ERP first with no BPR?	The company may automate an inefficient existing process.
Best approach	Start with business goals and As-Is problems, then design the To-Be process using ERP best practices.

Recommended flow

Business Goal	As-Is	Problems	ERP Best Practice	To-Be	Configure

Best one-line conclusion: Do not automate a bad process, but do not redesign a process while ignoring what the ERP can already support.

Two related exam nuances

Topic	Revision point
ERP is a downsizing tool?	No. ERP may automate manual work and change roles, but its purpose is integration and business improvement.
Multi-level hierarchy / Subway case	Hierarchy itself is not failure. The problem is unnecessary hierarchy that slows end-to-end processes and decisions.

3 ERP Readiness, Success Factors and Risks

Readiness Audit: check whether the organization is ready before implementation. The course notes explicitly identify Readiness Audit and Performance Measurement as ERP truths.

Readiness area	Quick check
Management	Is leadership committed and clear about the business objective?
Process	Are core processes understood well enough to redesign/configure?
Data	Is master data accurate and usable?
People	Will users accept the change and receive training?
Resources	Are budget, time and implementation capability realistic?

Success factors - remember the logic, not a long list

Management support + good processes + clean data + trained users + change acceptance + realistic goals + suitable ERP + disciplined implementation.

Risk framework

Risk type	Typical problem
People	Resistance, poor training, weak ownership.
Process	Bad process gets automated; process fit is poor.
Data	Wrong master data creates wrong planning and reporting.
Technology	Integration, performance, security or stability issues.
Project	Scope creep, delays, cost overruns and weak governance.

Memory: People - Process - Data - Technology - Project.

Performance Measurement

After go-live, judge success through business outcomes such as order cycle time, inventory, on-time delivery, stock-outs, error rates and process lead time - not merely whether the software is running.

4 Evaluate ERP as a System: the 5 Cs

Your notes provide five evaluation dimensions: **Complete, Connected, Cognitive, Compliant, Capable.**

C	Meaning	Question to ask
Complete	Supports complete business processes.	Does the system cover the end-to-end process?
Connected	Links internal and external stakeholders/systems.	Can departments, suppliers and customers connect?
Cognitive	Supports pattern detection / intelligent insight.	Can the system detect useful patterns or failures?
Compliant	Supports legal, SOP, GMP and best-practice needs.	Can we operate within required rules?
Capable	Handles required speed and transaction volume.	Can it handle the scale reliably?

Fast evaluation formula: Does it cover the process? Connect everything? Generate insight? Maintain compliance? Handle scale?

Quick myths to reject

Myth	Correct revision point
ERP = only software	ERP is a business-management approach enabled by integrated technology.
ERP = only manufacturing	ERP supports multiple business functions and service organizations too.
ERP solves every business problem	ERP provides visibility/integration; management must still improve processes.
More ERP systems = more productivity	Fragmentation can increase complexity; integration matters.
ERP = downsizing	Job redesign may occur, but downsizing is not the core purpose.

5 ERP Implementation Approaches

The notes explicitly list three approaches: **Big Bang, Franchising and Slam-dunk.**

Approach	Core idea	Main advantage	Main risk / limitation
Big Bang	Most of the organization switches at roughly the same time.	Fast transition.	High go-live risk if major problems occur.
Franchising	Roll out by site/unit/division with a common core and local differences.	Useful for dispersed or semi-independent units.	Needs strong coordination and standards.
Slam-dunk	Implement a limited, clearly defined scope instead of everything.	Focused and faster value.	Does not create full-enterprise transformation immediately.

Memory: Big Bang = all at once. Franchising = unit/site-wise. Slam-dunk = limited focused scope.

Pan■■nga / Five Elements - not a new topic

The course slide names an ERP Pan■■nga concept, but the detailed five-element explanation is already covered in your implementation-methodology notes: Client, User, ERP Brand, Consultant and Methodology. So do not treat it as an extra chapter.

6 Value Matrix Analysis

The lecture slide is a photographed whiteboard. The clearly visible conceptual progression is:

Production	**Cost Reduction**	**Customer**	**Service**
SIC	**MRP**	**MRP-II**	**ERP**

Stage	Main business concern
Production	Can we produce enough?
Cost Reduction	Can we control material and operating cost better?
Customer	Can we respond better to demand and customer needs?
Service	Can the whole enterprise deliver integrated customer value?

Core idea: As competitive priorities changed, the role of planning/information systems also evolved. Remember the direction, not every mark on the whiteboard.

Source caution: This is a compact interpretation of the visible lecture diagram; the slide itself does not provide a written explanatory paragraph.

7 ERP / Vendor Selection

Several FAQ lines about vendors, vendor strengths and ERP selection can be answered with one framework.

Selection framework: FIT - COST - SCALE - SUPPORT - INTEGRATION

Criterion	What to ask
Fit	Does the ERP fit our industry, processes and business model?
Cost	What is the total cost: license/subscription, implementation, training and maintenance?
Scale	Can it support future growth, users, sites and transaction volume?
Support	Is the vendor/implementation-partner ecosystem strong enough?
Integration	Can it connect with our existing and future systems?
Customization	Can we use standard best practices instead of heavily customizing the ERP?

Vendor examples present in the course slide

Examples include SAP, Oracle, Microsoft Dynamics 365, Infor, Epicor, IFS, NetSuite, Odoo, QAD, Sage, Workday, Plex and others.

Important: The uploaded vendor slide lists names, but does not state their strengths. So for exam revision, prioritize the selection framework over memorizing unsupported vendor-strength claims.

8 Master Data, Item Types and Signal Code

Master Data

Definition: relatively stable core information reused across many transactions.

Master data example	Used for
Customer Master	Customer code, address, payment terms and other reusable customer details.
Supplier Master	Supplier identity, terms and reusable purchasing information.
Item Master	Item code, description, unit, classification and planning-related information.
Employee Master	Reusable employee information.
Chart of Accounts	Standard financial account structure.

Master Data = who/what. Transaction Data = what happened.
Example: Item Master = Motor X. Transaction = purchased 50 Motor X today.

Item Types

Conceptual expansion used in our study: item type tells ERP what kind of item it is and therefore how the system should treat it. Useful examples: Purchased, Manufactured, Cost, Service and Generic.

Item Type	Simple meaning
Purchased	Bought from a supplier.
Manufactured	Produced internally.
Cost	Represents a non-physical cost.
Service	Represents a service rather than stocked physical inventory.
Generic	Used for configurable/product-variant situations.

Signal Code

Course-file limitation: the FAQ names Signal Code but does not define it. In our study, we treated it as a warning/status flag attached to an item so the ERP can alert the user to a special condition such as obsolete, blocked or invalid.

Easy distinction: Item Type = what the item is. Signal Code = something special you should notice about the item.

9 60-Second Cram Sheet

Topic	One-line recall
Modules	Different functions, one integrated ERP.
Manufacturing	Demand -> MPS -> BOM -> MRP -> Capacity -> Shop Floor.
BPR + ERP	Fix the process, then configure ERP around the improved To-Be process.
Readiness	Check management, process, data, people and resources before go-live.
Risks	People - Process - Data - Technology - Project.
5 Cs	Complete - Connected - Cognitive - Compliant - Capable.
Big Bang	Everything at roughly the same time.
Franchising	Site/unit-wise rollout.
Slam-dunk	Limited focused implementation.
Value Matrix	Production -> Cost -> Customer -> Service; SIC -> MRP -> MRP-II -> ERP.
ERP Selection	Fit - Cost - Scale - Support - Integration.
Master Data	Stable reusable core records.
Item Type	What the item is.
Signal Code	Special warning/status flag about an item.
Hierarchy	Hierarchy is fine if it does not create unnecessary delay.
Downsizing myth	ERP aims at integration and improvement, not simply reducing headcount.

Revision strategy: Read this page first, then revisit only the sections where a term does not immediately trigger the meaning in your head.

This Part 4 PDF intentionally avoids repeating the detailed material already captured in Parts 1-3.